ALISON SEONGJEE PARK

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RESEARCH INTERESTS

multi-omics data integration, deep learning/machine learning applications in precision oncology, single-cell genomics, cancer immunology, and disease subtype classification, networks

EDUCATION

CUNY School of Medicine | New York, NY

MD Candidate | August 2021 - January 2023

Sophie Davis | The City College of New York | New York, NY

BS Biomedical Sciences, Cum Laude | August 2017 - May 2021

Flatiron School | New York, NY

Data Science Bootcamp | February 2023 - June 2023

RESEARCH EXPERIENCE

Bioinformatician

Icahn School of Medicine at Mount Sinai | Multiple Myeloma Computational Research | December 2023 - Present

- Integrated multi-omics data (bulk RNA-seq, single-cell RNA-seq, clinical parameters) using unsupervised machine learning approaches (SNF, cNMF, MOVICS, MOFA, integrAO) to stratify newly diagnosed multiple myeloma patients, identifying immune-based disease subtypes with distinct tumor microenvironment signatures and prognostic implications
- Developed patient similarity networks through multi-omics integration to refine multiple myeloma disease classification beyond traditional staging systems, revealing novel molecular subtypes associated with differential outcomes
- Performed comprehensive single-cell RNA-seq analysis (Scanpy, pySCENIC) comparing tumor microenvironments between high-risk and standard-risk patients, identifying depleted antigen-presenting cell and IFN-stimulated CD4 T cell populations in aggressive disease
- Conducted regulon activity analysis using pySCENIC to identify transcription factor networks differentially active between risk groups, analyzing regulatory programs across multiple cell type subclusters to understand disease heterogeneity
- Managed and analyzed large-scale genomic datasets from multi-institutional cohorts, implementing reproducible computational pipelines and developing custom analysis solutions when existing tools proved insufficient for complex integration tasks
- Collaborated with clinical researchers and oncologists to translate computational findings into biological insights, contributing to 9 publications (3 manuscripts, 1 under review, 4 ASH conference abstracts with 2 awards)

Research Assistant

Icahn School of Medicine at Mount Sinai | Cancer Epigenetics Laboratory | August 2018 - November 2021

- Investigated epigenetic regulators (SETD8, E2F1) as therapeutic targets in multiple myeloma and mantle cell lymphoma using in vitro and in vivo models, contributing to drug discovery efforts that identified a potent SETD8 inhibitor advancing to preclinical development
- Analyzed complex experimental datasets using statistical methods (t-tests, ANOVA, survival analysis) and data visualization tools (GraphPad Prism, R), contributing to 2 presentations at ASH Annual Meeting
- Collaborated in cross-functional research teams to design experiments, optimize protocols, and troubleshoot technical challenges, directly contributing to 2 peer-reviewed publications including a manuscript under review in Journal of Medicinal Chemistry

Research Trainee

Icahn School of Medicine at Mount Sinai | Cancer Biology Laboratory | August 2017 - August 2018

- Trained in translational cancer biology research investigating epigenetic mechanisms in hematologic malignancies, performing molecular biology experiments and contributing to multiple research projects

Research Intern

Icahn School of Medicine at Mount Sinai | Tisch Cancer Institute | June 2016 - August 2016

- Trained in basic and translational cancer biology research in chemo-resistant muscle invasive bladder cancer, performing mouse surgeries, immunohistochemistry, nucleic acid extraction, PCR, and genotyping

Research Intern

Feinstein Institute for Medical Research, Northwell Health | June 2015 - August 2015

- Trained in cancer biology research investigating chronic lymphocytic leukemia, performing cell culture, Western blots, and flow cytometry

TECHNICAL SKILLS

Programming & Data Science

Python (scikit-learn, NumPy, Pandas, Keras, PyTorch, TensorFlow, Scanpy, AnnData), R (Bioconductor, Seurat, ggplot2, DESeq2), SQL, Git/GitHub, Shell scripting

Machine Learning & Computational Methods

Deep learning (CNNs, RNNs, autoencoders), multi-omics integration (SNF, cNMF, MOVICS, MOFA, integrAO), gradient boosting (XGBoost, LightGBM), dimensionality reduction (PCA, UMAP, t-SNE), feature engineering, clustering algorithms, statistical modeling, survival analysis

Bioinformatics & Genomics

Single-cell RNA-seq analysis (Scanpy, Seurat, pySCENIC), bulk RNA-seq (STAR, DESeq2, limma, GSEA), transcription factor regulon analysis, whole genome sequencing, variant calling, genomics pipeline development (Nextflow), network analysis, pathway enrichment

Cloud Computing & Infrastructure

High-performance computing (HPC), workflow management systems, Docker, version control, reproducible research practices

Laboratory Techniques

PCR/qPCR, Western blotting, cell culture, immunohistochemistry, flow cytometry, mouse xenografts, mouse surgeries, nucleic acid/protein extraction, experimental design & troubleshooting

SELECTED PUBLICATIONS & PRESENTATIONS

Peer-Reviewed Publications

Accepted - Chen, H., Dutta, R., Li, Z., Zhong, Y., Ma, A., Park, K-S., Kottur, J., **Park, A.**, Babault, N., Wang, K., Wang, D., Xiong, Y., Kaniskan, H., Luo, M., Parekh, S., Jin, J. Discovery of a Potent, Selective and In Vivo Efficacious Covalent Inhibitor for Lysine Methyltransferase SETD8. *Journal of Medicinal Chemistry*.

Conference Presentations & Abstracts

Hamidi, H., **Park, A.**, Dos Santos, J. V., Diaz, G., Okholm, T. L., Quidwai, M., ... & Lagana, A. (2025). Integrating Microenvironment with Tumor Multi-Omic using Unsupervised Machine Learning to Model Heterogeneity Refines Multiple Myeloma Subtypes and Reveals Immune-Based Clusters with Prognostic Impact. *Clinical Lymphoma Myeloma and Leukemia*, 25, S181-S182.

Vieira, J., **Park, A.S.**, et al. (2024). Immune Profiling Reveals Distinct Features of Multiple Myeloma Subtypes Defined By Multi-Omics Network Analysis. *Blood*, 144 (Supplement 1): 4658. [ASH Annual Meeting] [**Award**]

Vieira, J., **Park, A.S.**, et al. (2024). Single-Cell Analysis Reveals Depletion of Antigen-Presenting Cell and IFN-Stimulated CD4 T Cell Populations in High-Risk Newly Diagnosed Multiple Myeloma Patients. *Blood*, 144 (Supplement 1): 4654. [ASH Annual Meeting] [**Award**]

Edwards, D.R., Kuo, P., Lagana, A., **Park, S.**, et al. (2020). Aberrant Cell Cycle Programming Confers Rapid Lethality in the EuSOX11+ CCND1 MCL Mouse Model. [ASH Annual Meeting 2020]

Lagana, A., **Park, S.**, et al. (2018). E2F1 Is a Biomarker of Selinexor Resistance in Relapsed/Refractory Multiple Myeloma Patients. [ASH Annual Meeting 2018]

HONORS & AWARDS

ASH Abstract Achievement Award (2 awards) | American Society of Hematology | 2024

Cum Laude | Sophie Davis School of Biomedical Sciences | May 2021

PROFESSIONAL DEVELOPMENT

ASH Annual Meeting Attendee | American Society of Hematology | 2018, 2020, 2024, 2025

Data Science Intensive Training | Flatiron School | 2023

ADDITIONAL EXPERIENCE

Organizer

Katalyst 2025 | Los Angeles, CA | October 2025 – Present

- Organize national conference connecting Korean American undergraduates with mentors and career pathways in STEM, business, and entrepreneurship
- Coordinate logistics, speaker recruitment, and workshop programming for 200+ attendees
- Build community infrastructure to support underrepresented students in pursuing advanced degrees and professional development Presenter

UKC FIRE 2025 | Atlanta, GA | August 2025

- Presented computational biology research on multiple myeloma patient stratification to interdisciplinary audience of scientists and industry professionals
- Received 2nd place for innovative research approach and translational potential
- Engaged with mentors across academia and biotechnology to refine research direction Brand Associate

ELOREA | New York, NY | August 2023 - September 2024

- Developed data-driven sales strategies and analytical frameworks for early-stage beauty startup
- Built customer segmentation models and performed market analysis to optimize product positioning
- Analyzed sales performance metrics and customer behavior patterns to inform business decisions